Iterative Time Complexity Examples (IRZA)

1.
```
algo sum(arr, n){
    total = 0
    for (int i = 1; i <= n; i++){
        total = total + arr[i]
    }
    return total
}
```

2.
```
algo sum(arr, n){
    total = 0
    for (int i = n; i >= 0; i--){
        total = total + arr[i]
    }
    return total
}
```

3.
```
for (int i = 0; i < n; i++){
    total = total + arr[i]
}
for (int j = 0; j < n; j++){
    total = total + brr[j]
}
```

4.
```
for (int x = 0; x < n; x = x + 2){
    break
}
```

5.
```
for (int i = 0; i < n; i = i + 3){
    print("X");
}
```

6.
```
for (int i = n; i > 0; i = i - 4){
    print("X");
}
```

7.
```
for (int i = 1; i < n; i = i * 3){
    print("X");
}
```

8.
```
for (int i = n; i > 1; i = i / 4){
    print("X");
}
```

9.
```
for (int i = 1; i * i <= n; i++){
    print("X");
}
```

10.
```
algo add(arr, brr, n){
    for (int x = 0; x < n; x++){
        for (int y = 0; y < n; y++){
            C[x, y] = arr[x, y] + brr[x, y]
        }
    }
}
```

11.
```
algo multiply(arr, brr, n){
    for (int i = 0; i < n; i++){
        for (int j = 0; j < n; j++){
            C[i, j] = 0
            for (int z = 0; z < n; z++){
                C[i, j] = arr[i, z] * brr[i, z] + C[i, j]
            }
        }
    }
}
```

12.
```
for (int i = 0; i < n; i++){
    for (int j = 1; j < n; j = j * 3){
        print("X");
    }
}
```

13.
```
for (int i = 0; i < n; i++){
    for (int j = 1; j < n; j = j * 2){
        if (j > 100){
            break
        }
    }
}
```

14.
```
y = n
while y > 0:
    y -= y
```

15.
```
z = n
while z > 1:
    z = z // 3
```

16.
```
k = 1
while k < n:
    k = k + k
```

17.
```
val = 0
for (j = 1; val < n; j = j + 1){
    val = val + j
}
```

18.
```
for (int a = 1; a <= n; a = a + 1){
    for (int b = 1; b <= a; b++){
        print()
    }
}
```

19.
```
for (int x = 1; x < n; x = x + 1){
    y = 1
    while (y * y <= x){
        y += 1
    }
}
```

20.
```
for (int i = 1; i <= n * n; i++){
    for (int j = 1; j <= i; j++){
        print()
    }
}
```

21.
```
int i, j, k, m, multi, a, b, c;
for (i = n; i >= 1; i = i / 5) {
    for (j = 1; j <= n; j = j + 2) {
        for (k = 1; k <= 50; k = k + 1) {
            multi = a * b;
        }
        for (m = 1; m <= n; m = m + 4) {
            multi = multi * c;
        }
    }
}
```

Time Complexity Solutions

1. $O(n)$
2. $O(n)$
3. $O(n)$
4. $O(1)$
5. $O(n)$
6. $O(n)$
7. $O(\log n)$
8. $O(\log n)$
9. $O(\sqrt{n})$
10. $O(n^2)$
11. $O(n^3)$
12. $O(n \log n)$
13. $O(n)$
14. $O(1)$
15. $O(\log n)$
16. $O(\log n)$
17. $O(\sqrt{n})$
18. $O(n^2)$
19. $O(n\sqrt{n})$
20. $O(n^4)$
21. $O(n^2 \log n)$